

Structured Cabling & Workspace Deployment

Cisco Packet Tracer Physical Mode Engineering Documentation

Lab Overview & Objectives

This implementation lab models structural data distribution and horizontal cabling standards within enterprise environments. Key components include the installation of data patch panels in an Equipment Closet, deployment of workspace Wall Mount blocks, structural drop management via geometric bend points, and end-to-end verification utilizing dynamic host address discovery and standard application layer inquiries.

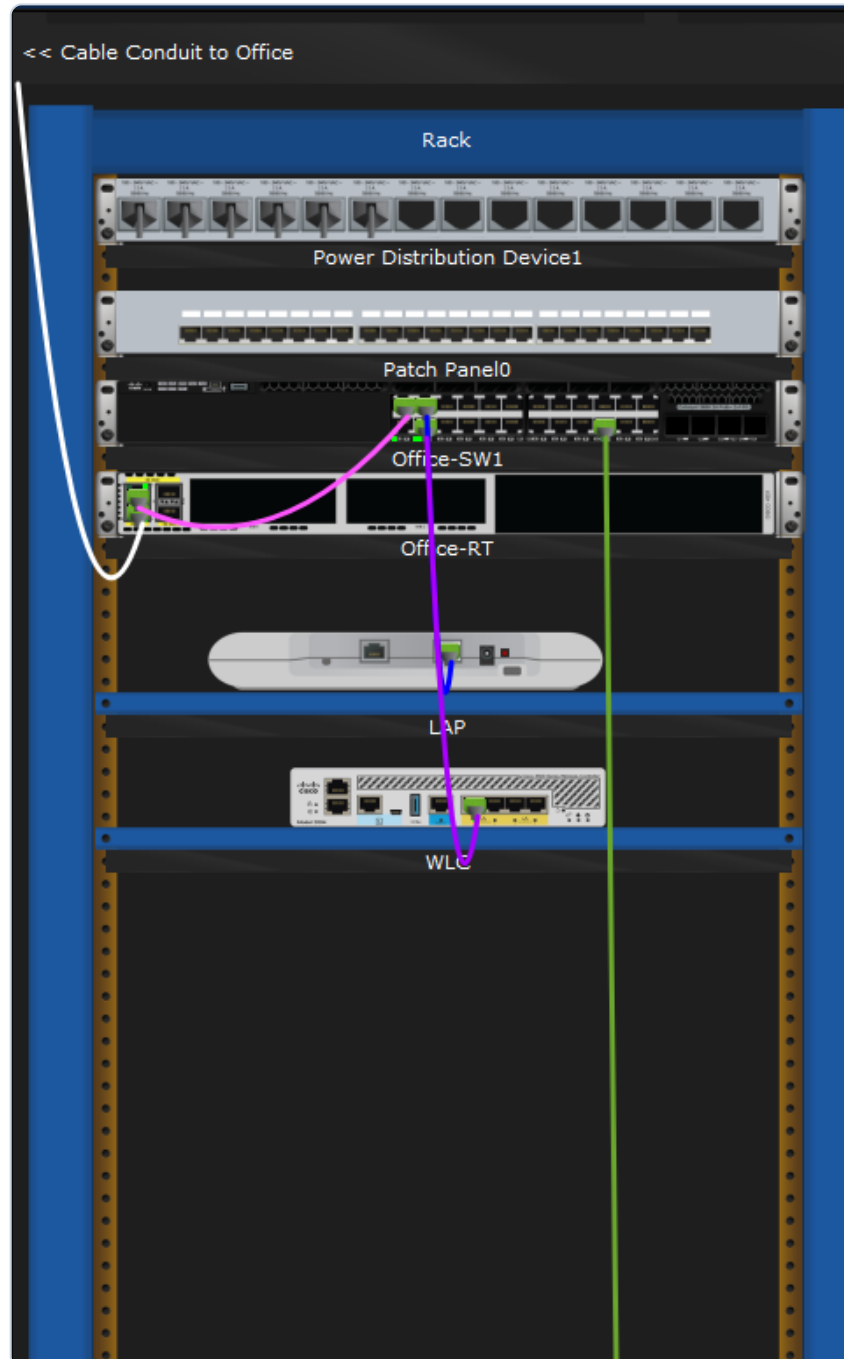
Part 1: Installing & Interconnecting Rack Patch Panels

Concept: Cross-Connect Architecture

A **Patch Panel** transitions institutional vertical backbone distribution down to structural horizontal distributions. Running intermediate cross-connections protect expensive operational networking hardware switches from physical operational deterioration caused by frequent drop alterations or testing activities.

Step 1: Equipment Closet Hardware Provisions

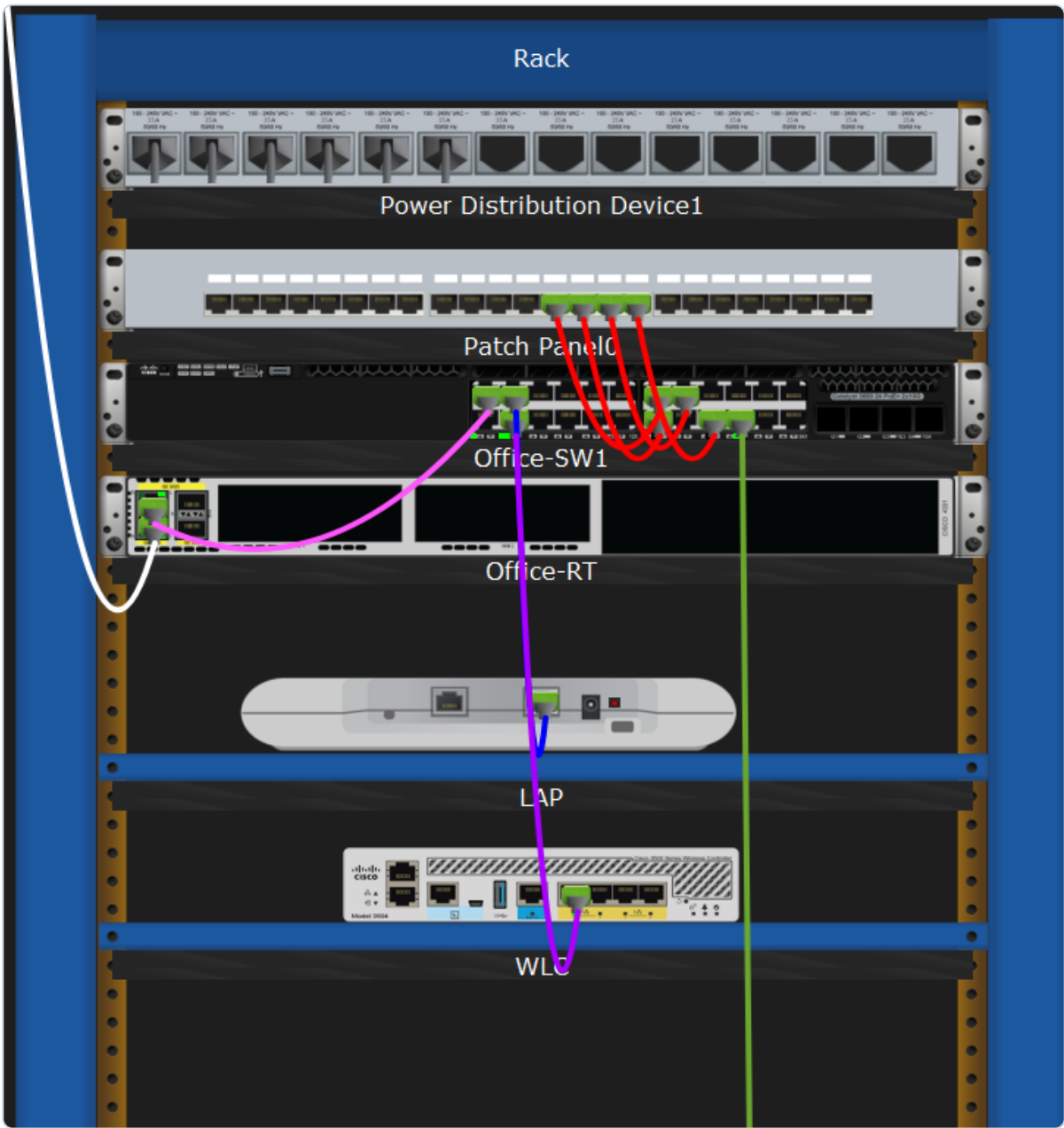
Accessed the simulated **Equipment Cabinet** closet, added a standardized **Copper Patch Panel** directly within the framework hierarchy, and confirmed system tagging as Patch Panel0.



Step 2: Switch Mapping & Cross-Connect Bundles

Deployed high-visibility Red Straight-Through cabling pathways connecting data ports directly across to dedicated distribution channels. Employed the software cable organizer to establish clean vertical drop layouts:

Source Switch Port (Office-SW1)	Destination Jack (Patch Panel0)
GigabitEthernet 1/0/13	Jack13
GigabitEthernet 1/0/14	Jack14
GigabitEthernet 1/0/15	Jack15
GigabitEthernet 1/0/16	Jack16



Part 2: Primary Office Wall Mount Deployment & Cable Dressing

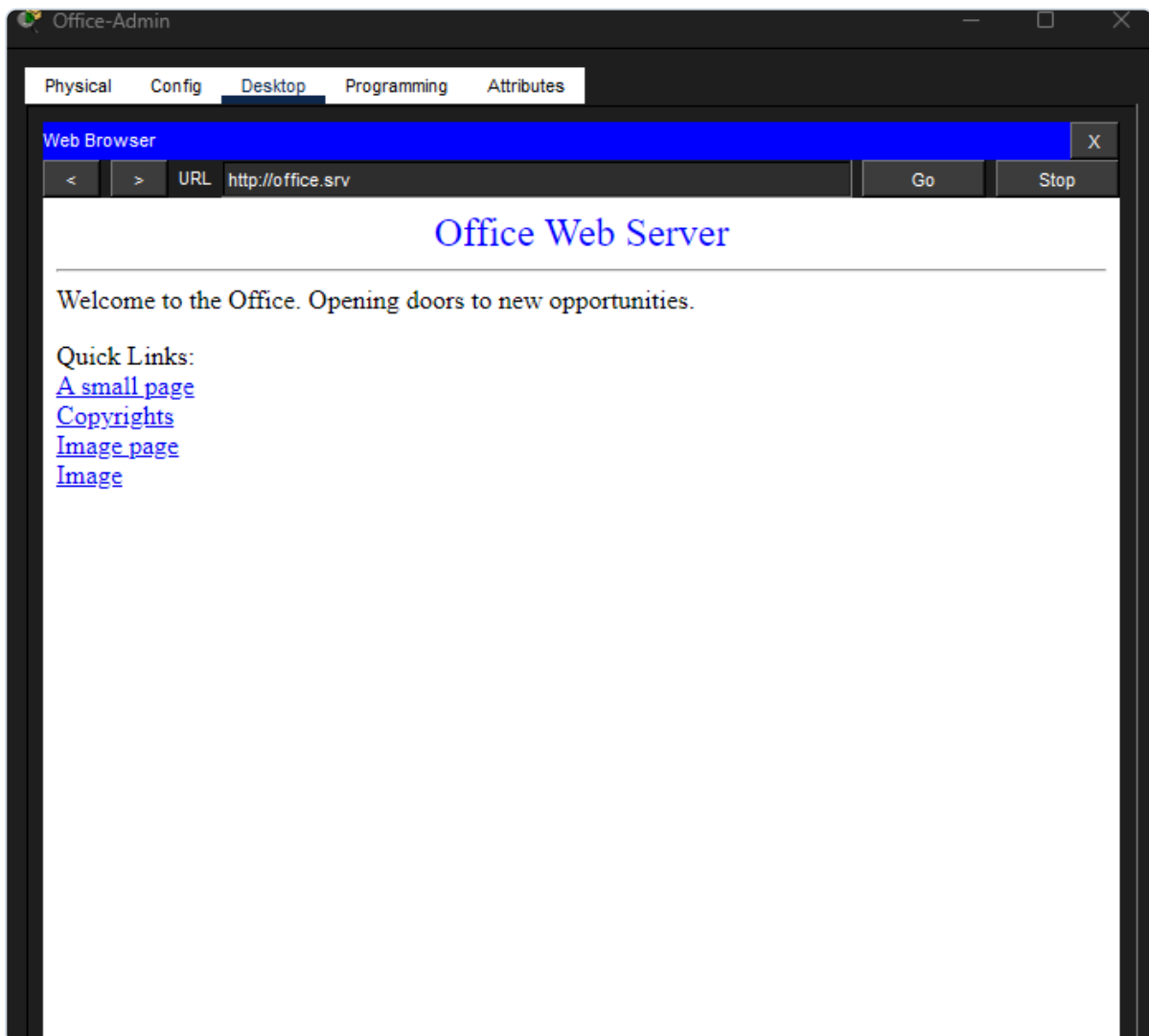
Concept: Horizontal Subsystem Drops

A **Wall Mount** acts as a physical termination breakout point in individual workspaces. Connection runs span out from back-end infrastructure IDC punchdowns within the rack to corresponding user RJ45 wall jack sockets, safely routing hidden structural cables inside conduits or walls.

Step 1: Structural Punchdowns & Administrative Endpoint Provisioning

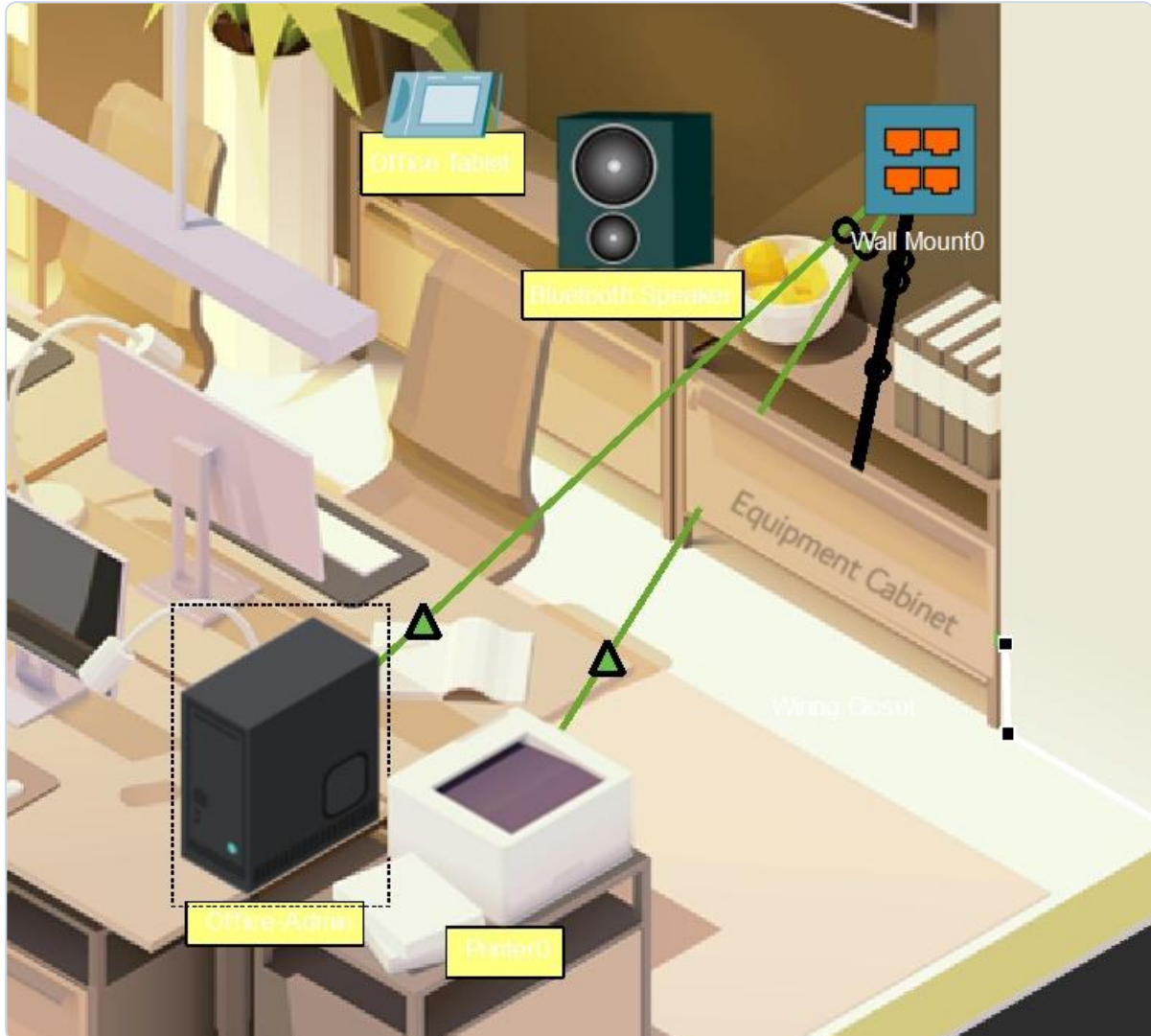
Attached Wall Mount0 alongside the primary enclosure cabinet. Punched down structural pairs traversing through the workspace envelope directly back into the core patch frame. Endpoints Office-Admin and Printer0 were then connected to the wall faceplates via standard patch leads.

Wall Mount0 Block Position	Core Patch Panel0 Punchdown Designation
Punchdown 1	Punchdown 13
Punchdown 2	Punchdown 14
Punchdown 3	Punchdown 15
Punchdown 4	Punchdown 16



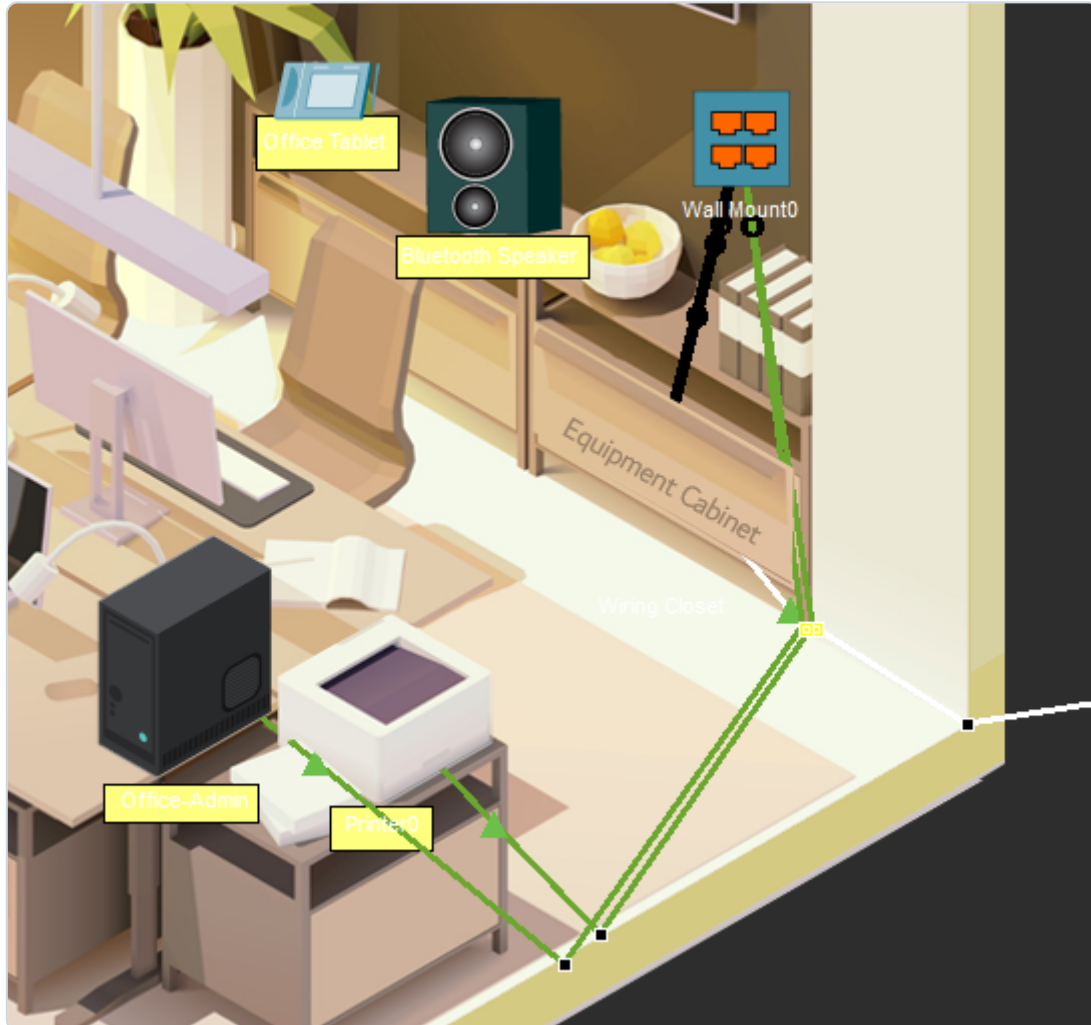
Step 2: Functional Core Verification

Once linked up, the stations successfully initialized standard parameters over DHCP from the local server. Verification checks were performed by launching a local terminal browser from Office-Admin and resolving the intranet portal site at <http://office.srv>.



Step 3: Geometrical Cable Routing Management

In physical enterprise spaces, dangling cables across paths introduce structural safety hazards. Handled loose cables by introducing strategic **BendPoints**, carefully anchoring paths along floorboards and perimeter walls to model proper indoor pathways.



Part 3: Expanding Infrastructure to Additional Workstations

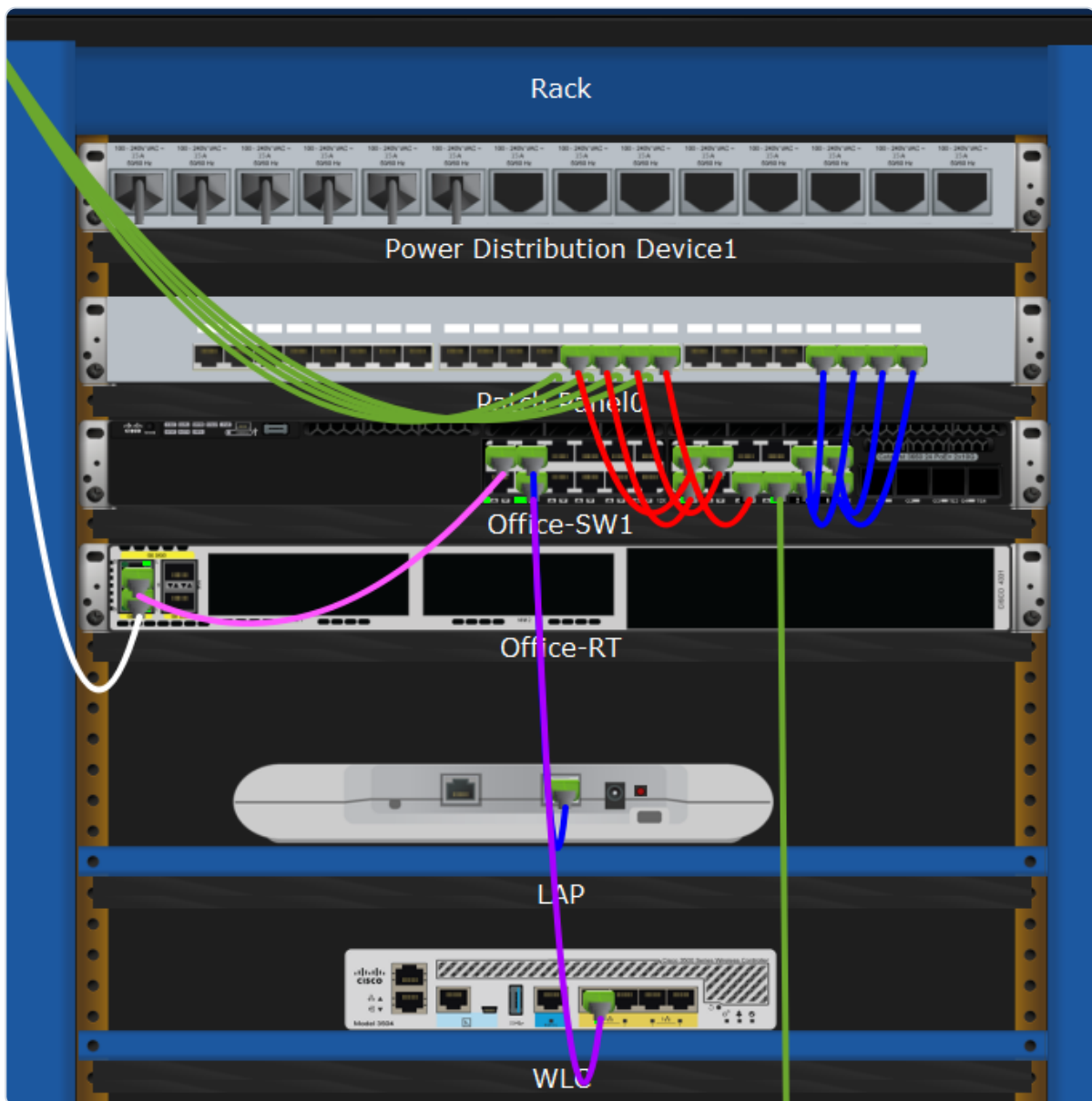
Concept: High-Density Distribution Layouts

Scaling operational branches requires provisioning additional access layers across the building topology. This expansion maps a secondary room location back to an independent distribution channel on the same central patch infrastructure.

Step 1: Secondary Block Wiring Configurations

Returned to the central equipment rack to configure a secondary block. Extended yellow-jacketed lines out from the active distribution switches to patch terminals:

Expansion Switch Interface (Office-SW1)	Central Patch Panel0 Assignment
GigabitEthernet 1/0/21	Jack21
GigabitEthernet 1/0/22	Jack22
GigabitEthernet 1/0/23	Jack23
GigabitEthernet 1/0/24	Jack24

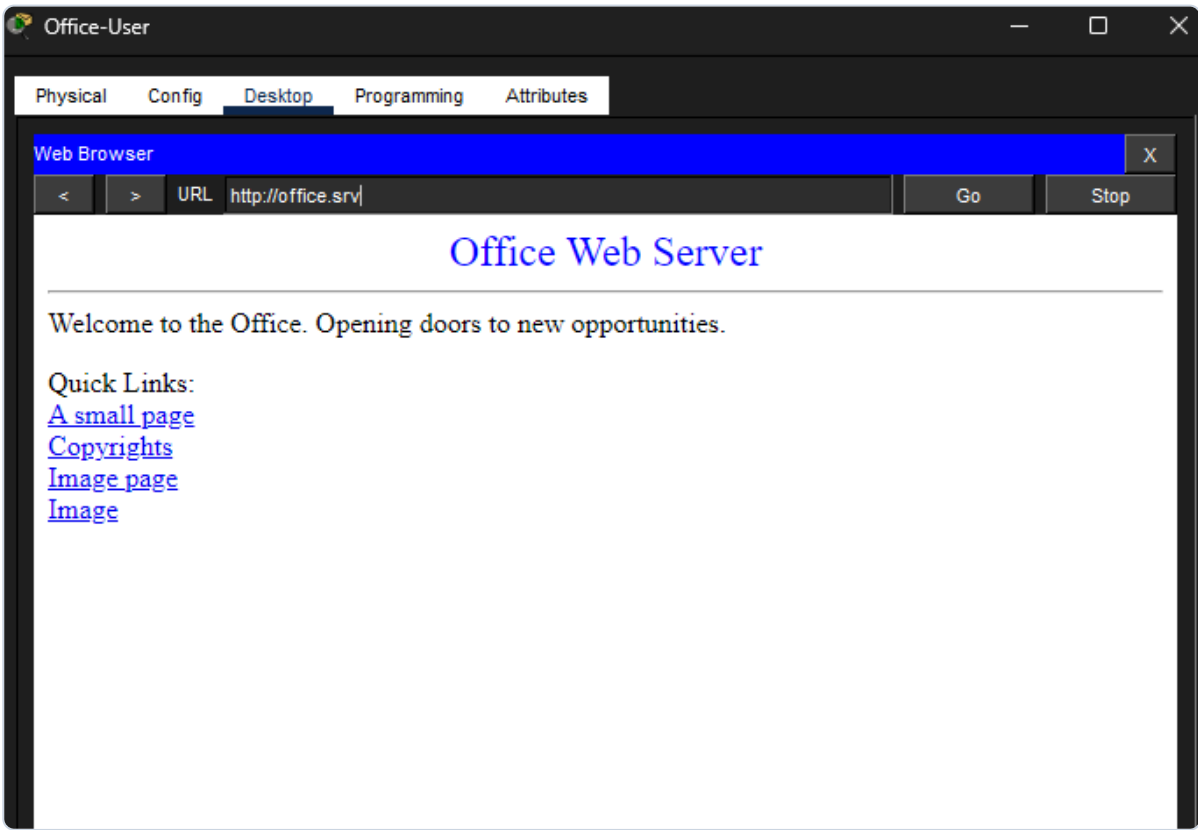


Step 2: Window-Side Workspace Deployment

Installed a distinct faceplate designated as Wall Mount1 right next to the workspace perimeter window. Structured cabling links were completed to connect the window drops back to the equipment frame:

Window Faceplate Outlet (Wall Mount1)	Central Patch Panel0 Punchdown Link
Punchdown 1	Punchdown 21
Punchdown 2	Punchdown 22
Punchdown 3	Punchdown 23
Punchdown 4	Punchdown 24

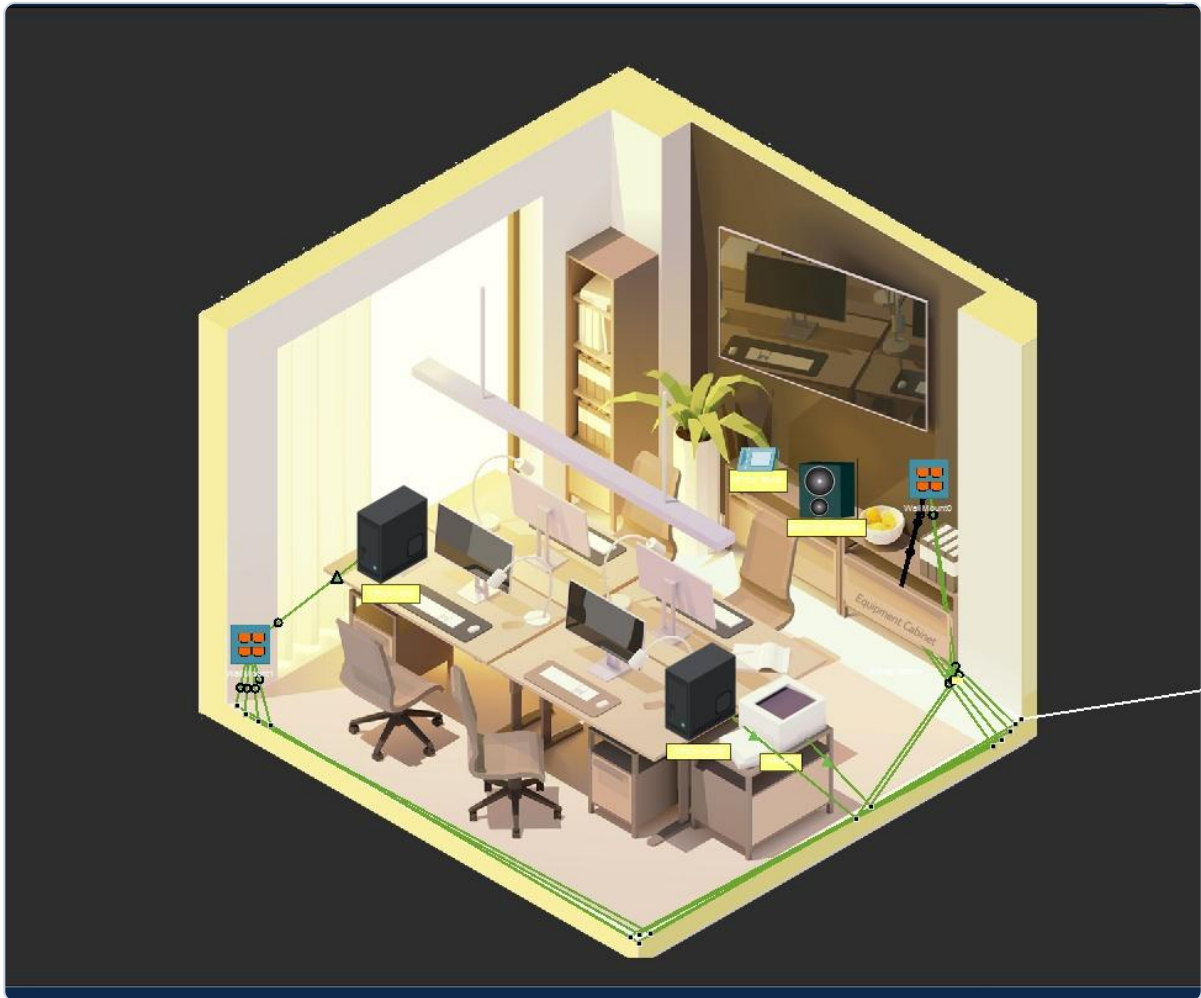
Connected the final Office-User station to the new window drop. Applied bend points to neatly tuck the long yellow cabling paths around structural office columns.



Part 4: Final Assessment & Verification Summary

Step 1: End-to-End Workspace Audits

Verified that the newly deployed Office-User PC completed full DHCP parameters initialization. Successfully loaded the web application server index layout page from the device desktop environment.



Step 2: Completed Workspace Overview

The complete physical floor configuration features well-managed horizontal lines, fully mapped communication ports, and clean pathways across all active business units.

Congratulations Guest! You completed the activity.

Overall Feedback **Assessment Items** Connectivity Tests

Expand/Collapse All Show Incorrect Items

Assessment Items	Status	Points	Component(s)	Feedback
- Network				
- Office-Admin				
- Default Gateway	Correct	1	Ip	
- Ports				
- FastEthernet0				
- IP Address	Correct	1	Ip	
- Link to Wall Mount0		0	Other	
- Type	Correct	1	Physical	
- Subnet Mask	Correct	1	Ip	
- Office-User		0	Other	
- FastEthernet0		0	Other	
- Link to Wall Mount1		0	Other	
- Type	Correct	1	Physical	
- Patch Panel0				
- Ports				
- Jack13		0	Other	
- Link to Office-SW1		0	Other	
- Connects to GigabitEthernet1/0/13	Correct	1	Physical	
- Jack14		0	Other	
- Link to Office-SW1		0	Other	
- Connects to GigabitEthernet1/0/14	Correct	1	Physical	
- Jack15		0	Other	
- Link to Office-SW1		0	Other	
- Connects to GigabitEthernet1/0/15	Correct	1	Physical	
- Jack16		0	Other	
- Link to Office-SW1		0	Other	
- Connects to GigabitEthernet1/0/16	Correct	1	Physical	
- Jack21		0	Other	
- Link to Office-SW1		0	Other	
- Connects to GigabitEthernet1/0/21	Correct	1	Physical	
- Jack22		0	Other	
- Link to Office-SW1		0	Other	
- Connects to GigabitEthernet1/0/22	Correct	1	Physical	
- Jack23		0	Other	
- Link to Office-SW1		0	Other	
- Connects to GigabitEthernet1/0/23	Correct	1	Physical	
- Jack24		0	Other	
- Link to Office-SW1		0	Other	
- Connects to GigabitEthernet1/0/24	Correct	1	Physical	
- PunchDown13		0	Other	
- Link to Wall Mount0		0	Other	
- Connects to PunchDown1	Correct	1	Physical	
- PunchDown14		0	Other	
- Link to Wall Mount0		0	Other	
- Connects to PunchDown2	Correct	1	Physical	
- PunchDown15		0	Other	
- Link to Wall Mount0		0	Other	
- Connects to PunchDown3	Correct	1	Physical	
- PunchDown16		0	Other	
- Link to Wall Mount0		0	Other	
- Connects to PunchDown4	Correct	1	Physical	
- PunchDown21		0	Other	
- Link to Wall Mount1		0	Other	
- Connects to PunchDown1	Correct	1	Physical	
- PunchDown22		0	Other	
- Link to Wall Mount1		0	Other	
- Connects to PunchDown2	Correct	1	Physical	
- PunchDown23		0	Other	
- Link to Wall Mount1		0	Other	
- Connects to PunchDown3	Correct	1	Physical	
- PunchDown24		0	Other	
- Link to Wall Mount1		0	Other	
- Connects to PunchDown4	Correct	1	Physical	

Step 3: Laboratory Task Completion Verification

The Packet Tracer engine verification routine recorded a 100% completion metrics profile, confirming that all physical mounts, patching configurations, color codes, and logical service checks meet target criteria.

- Printer0	0	Other
- Ports	0	Other
- FastEthernet0	0	Other
- Link to Wall Mount0	0	Other
- Type	Correct	1
		Physical